

Assignment 1: Vehicle Modeling and Simulation

Autonomous Driving Systems - Planning and Control Module

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Introduction

This report presents the implementation and simulation of three vehicle models: Kinematic, Linear Single-Track, and Nonlinear Single-Track. The objective is to analyze vehicle dynamics under different maneuvers and compare the results obtained from different modeling approaches and numerical integration methods (Euler vs. RK4).

The parameters used for the simulations (mass, geometry, tire coefficients, aerodynamics) follow the assignment specifications.

1 Exercise 1: Vehicle Modeling and Simulation

In this exercise, the three models have been tested with sinusoidal steering input ($\delta_{max} = 0.1$ rad, $f = 0.5$ Hz) and constant longitudinal acceleration ($a_x = 1.0$ m/s²) at two different initial velocities (10 m/s and 27 m/s). The simulation has been run for a duration of 5 seconds with a time step of 0.001 seconds.

1.1 Cause of difference in slip angle

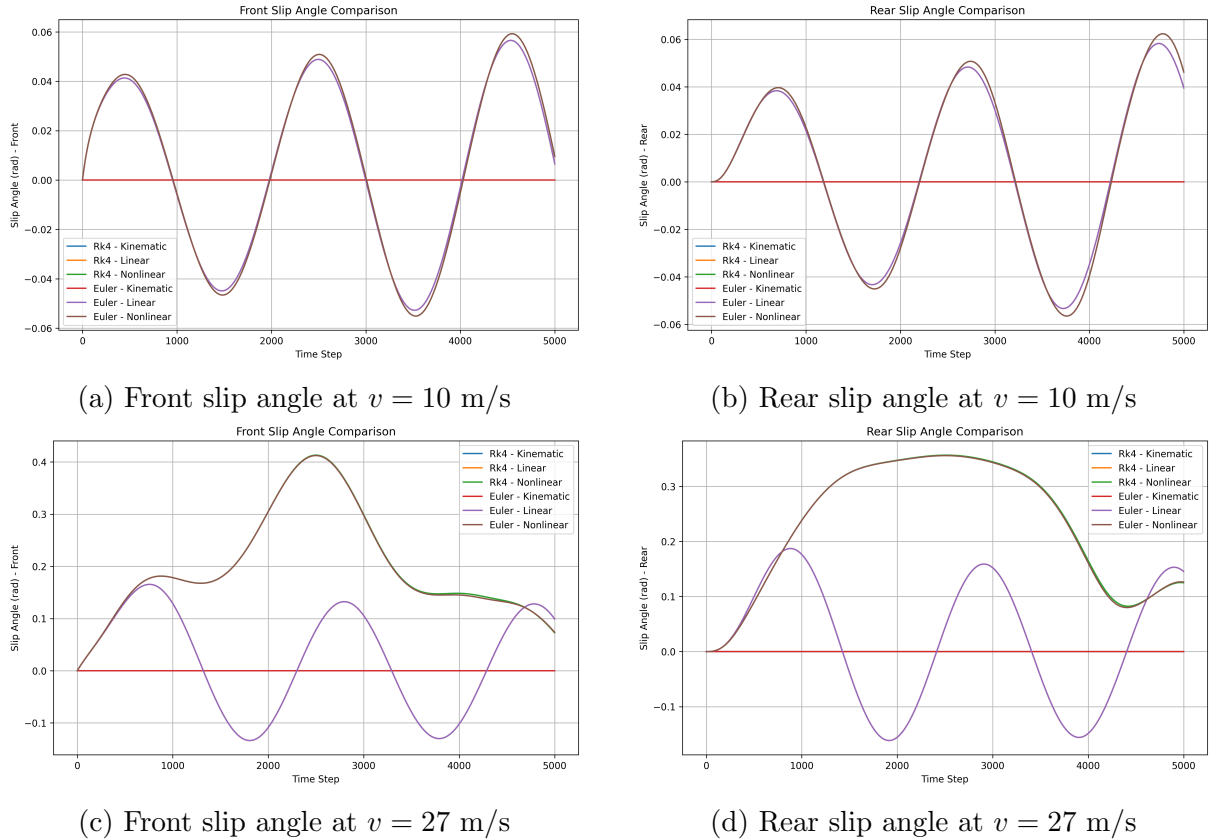


Figure 1: Slip angle trends at different velocities

A rather high difference is observed between the slip angle values from the linear and non-linear models at $v = 27$ m/s (Fig. 1c, 4d). Conversely, at $v = 10$ m/s the differences between the two models are negligible (Fig. 4c, 4d). This behavior is due to the different methods used to calculate the slip angle.

$$\alpha_f = \delta - \arctan\left(\frac{v_y + l_f r}{v_x}\right) \quad (1a) \qquad \alpha_r = -\arctan\left(\frac{v_y - l_r r}{v_x}\right) \quad (1b)$$

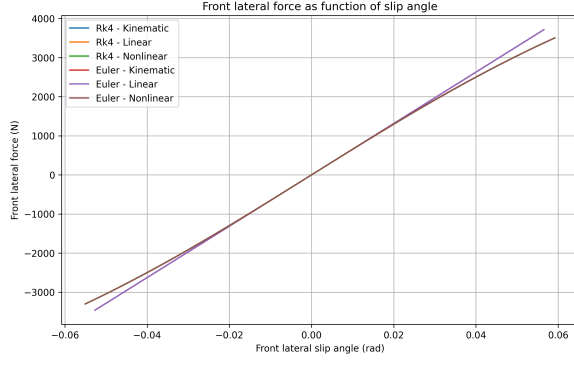
Figure 2: Nonlinear Single-Track Model slip angles

$$\alpha_f = \delta - \frac{v_y + l_f r}{v_x} \quad (2a) \qquad \alpha_r = -\frac{v_y - l_r r}{v_x} \quad (2b)$$

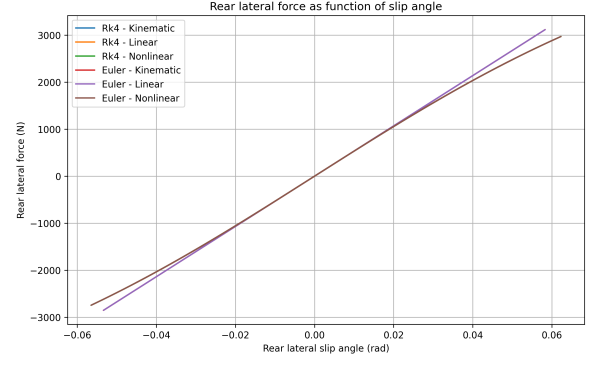
Figure 3: Linear Single-Track Model slip angles

As shown in Figure 3, the linear model replaces the nonlinear *arctan* term with its linear approximation. This is the reason why the model becomes, in fact, linear. However, this model has limited applicability, as the linear approximation of the *arctan* function is valid only for driving scenarios characterized by small slip angles. This typically implies a low lateral velocity v_y and a yaw rate r relative to the longitudinal speed v_x . In this specific case, the approximation remains sufficiently accurate and, in any event, introduces only a negligible error.

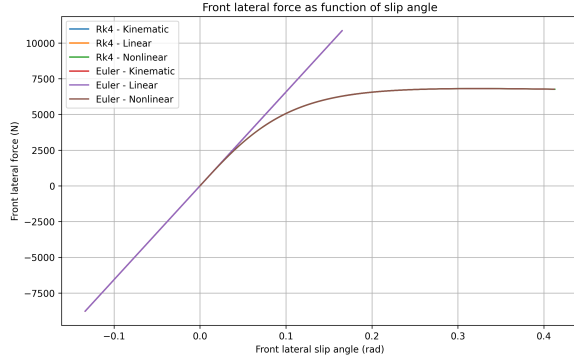
Both the examined scenarios are characterized by a sinusoidal maximum steering input $\delta_{max} = 0.1$ that, in combination with a low speed, allows to represent the vehicle behavior through the linear model; this is the reason why Figure 4c and Figure 4d show almost no differences between the slip angle values. However, the linear model does not hold for higher speed, since the *arctan* approximation is no more valid. This error is shown by Figure 1c and Figure 1d.



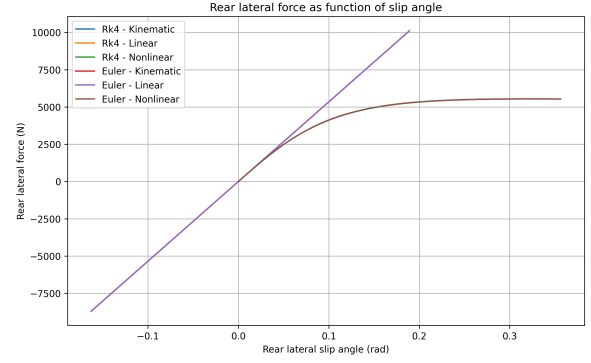
(a) Front lateral force vs slip angle at $v = 10$ m/s



(b) Rear lateral force vs slip angle at $v = 10$ m/s



(c) Front lateral force vs slip angle at $v = 27$ m/s



(d) Rear lateral force vs slip angle at $v = 27$ m/s

Figure 4: Lateral force vs slip angle

An additional consequence of the slip angle approximation is illustrated in Figure 4. The figure demonstrates that the lateral force F_y predicted by the linear tire model increases without bound beyond a certain vehicle speed, which is physically implausible. This divergence indicates the speed range beyond which the linear model is no longer valid. This effect arises because the lateral force F_y predicted by the linear model is directly proportional to the slip angle α (see 3) and therefore does not capture the saturation of the lateral force at higher slip angles. In contrast, the nonlinear model incorporates this saturation behavior through the use of the Magic Formula (see 4).

$$F_y = C_\alpha \cdot \alpha \quad (3)$$

Equation 3: Linear tire model.

$$F_y(\alpha) = F_N \cdot D \sin(C \arctan(B\alpha - E(B\alpha - \arctan(B\alpha)))) \quad (4)$$

Equation 4: Nonlinear tire model (Pacejka Magic Formula).

2 Exercise 2: Constant Steering and Acceleration

This exercise compares the models under constant steering inputs. For this exercise, the models have been tested with two different input steering angle ($\delta = 0.01$ rad and $\delta = 0.055$ rad). Both tests have been run with an initial speed $v_0 = 24$ m/s.

2.1 Significant differences and causes

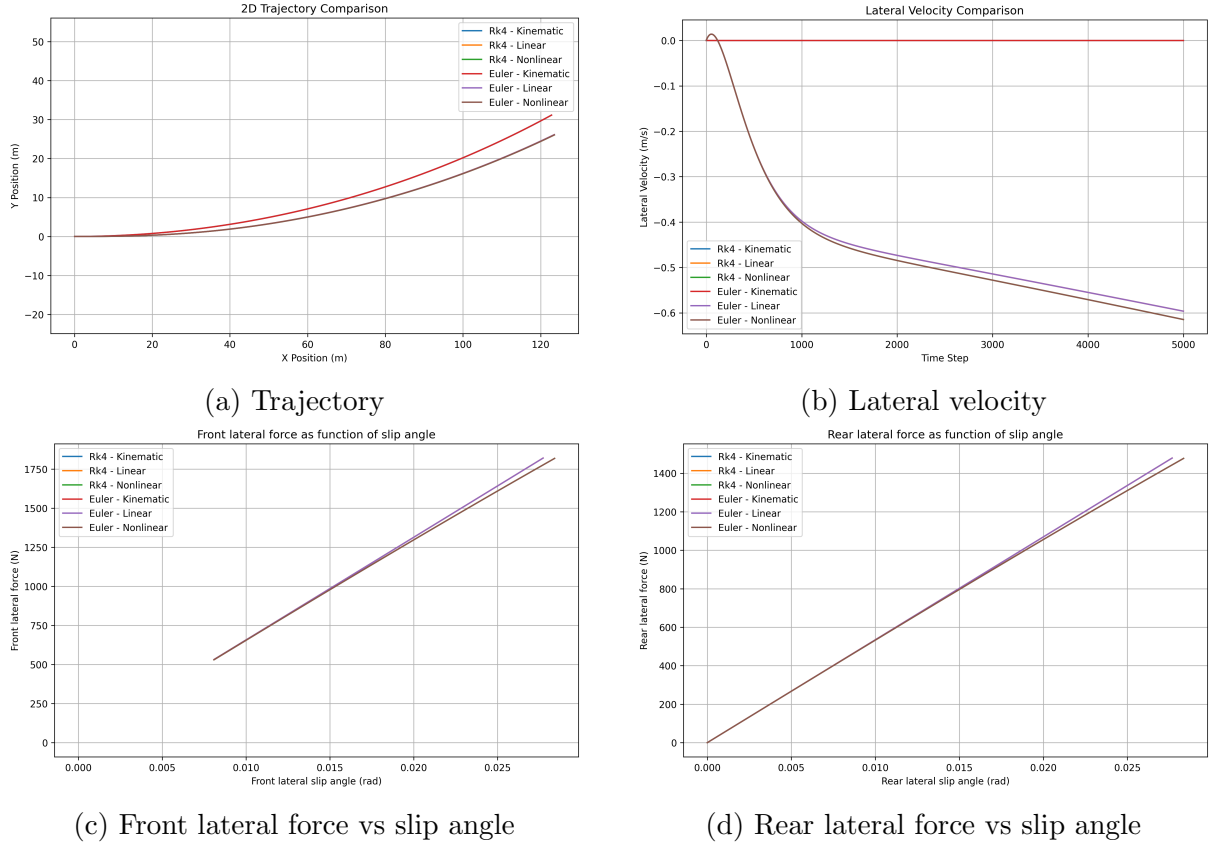


Figure 5: Test results with $\delta = 0.001$ rad

By running this test, a clear effect of what has been reported in the Section 1 can be observed. With a small steering angle ($\delta = 0.01$ rad) the generated slip angle remains small, therefore the linear model approximation is valid (Fig. 5c, 5d). Consequently, all trajectories are nearly identical (Fig. 5a). Conversely, with a larger steering angle, the slip angles become significant, rendering the linear model approximation invalid.

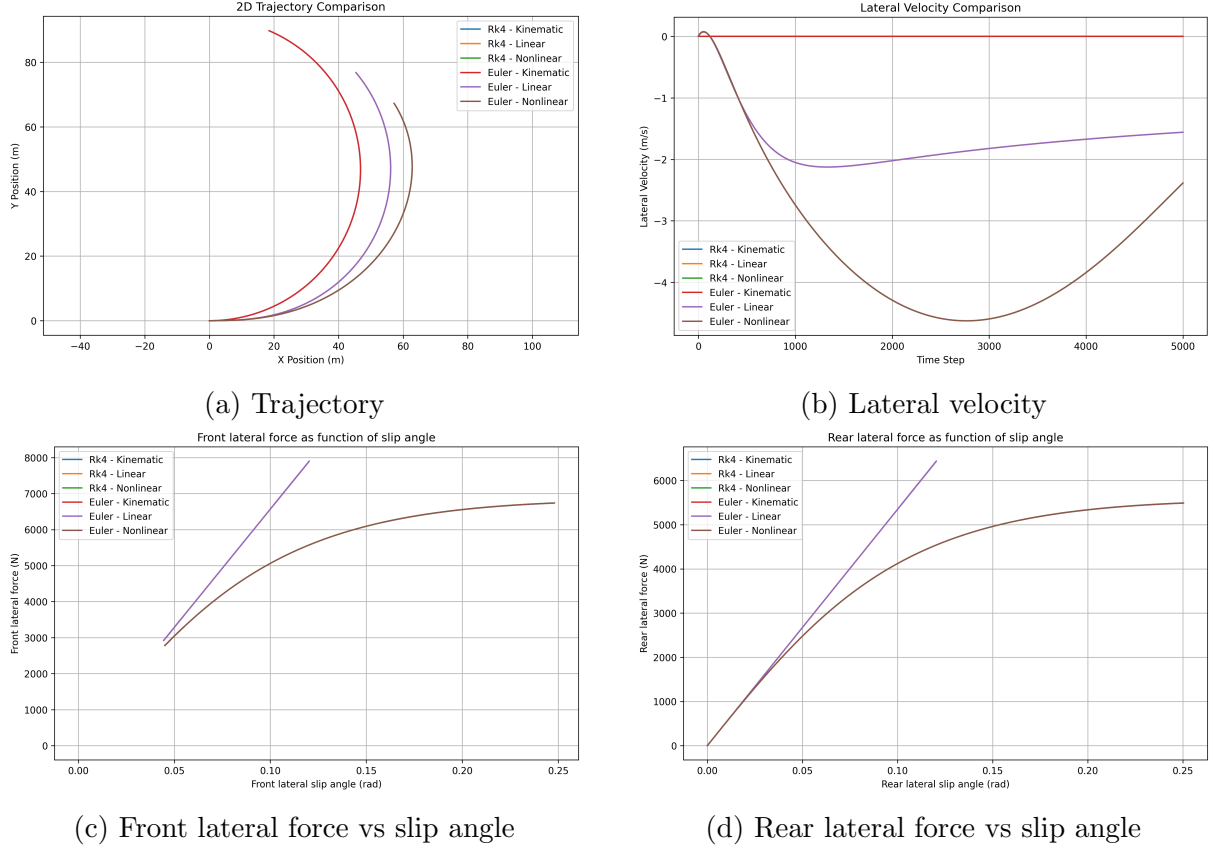


Figure 6: Test results with $\delta = 0.055$ rad

Specifically, the linear model predicts an unrealistic lateral force F_y value, due to the omission of the *arctan*. As previously said, the linear model doesn't account for the tire lateral force saturation, instead considered by the nonlinear model, resulting in the tires producing a lateral force F_y directly proportional to the slip angle α . On the other hand, the nonlinear model reaches the lateral force saturation value, calculated via the Magic Formula (Fig. 6c, 6d), meaning that the vehicle drifts and lose longitudinal velocity v_x in favor of lateral velocity v_y (Fig 6b) which did not happen in the previous case (Fig. 5b). This is the reason why the nonlinear model predicts a larger curve than the linear model. (Fig 6a)

3 Exercise 3: Comparing Numerical Integration Methods

This exercise aims to make a comparison between Euler and Runge-Kutta 4 (RK4) integration methods, executing the maneuver made in the Exercise 2 (Section 2, $\delta = 0.055$ rad) with a larger time step ($\Delta t = 0.08$ s) over a duration of 10 seconds.

3.1 Difference between Euler and RK4

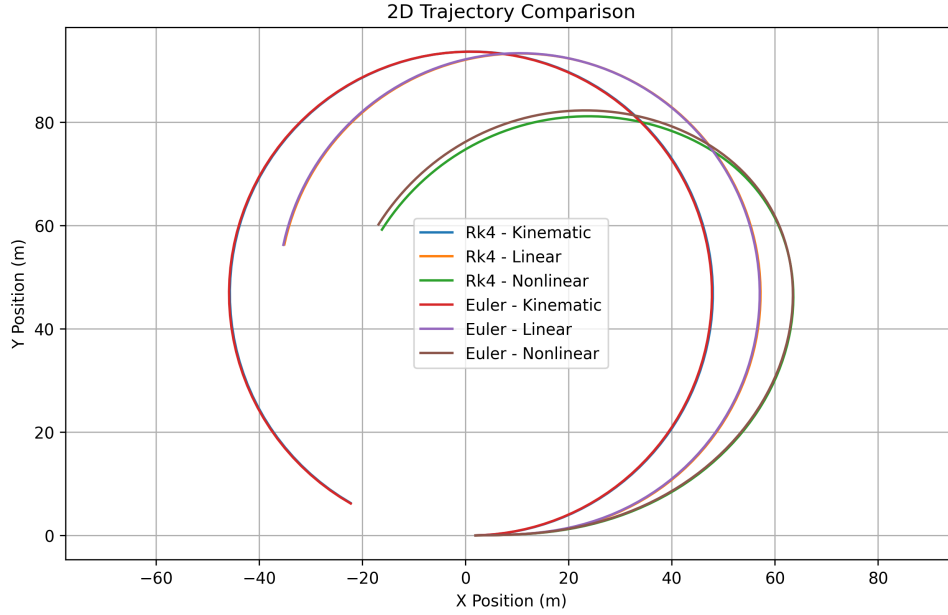


Figure 7: Trajectory comparison: Euler vs RK4 ($\Delta t = 0.08$ s)

The figure 7 shows a slightly different trajectories predicted by the nonlinear method: this is due to the two different used integration methods.

The Euler integration method is a first-order numerical integrator, meaning that it evaluates the derivative only once within each time step and then updates the state by adding to the previous value the increment obtained from this derivative evaluation. In practical terms, this corresponds to approximating the solution locally by a straight line connecting the state at the current time step to the estimated state at the subsequent time step. Because of this linear approximation, truncation errors accumulate over successive steps, which can lead to substantial global error and, consequently, an inaccurate approximation of the true trajectory.

RK4 is instead a fourth-order integrator, which calculates the slope at four different points within each time step and then computes a weighted average of these values. This makes it suitable for integration even with larger time steps, providing a more accurate solution than Euler's method.

It is noteworthy that trajectory differences are only visible in the nonlinear model. Euler's method is sufficient for integrating linear dynamics, but struggles with nonlinear functions due to their unpredictable curvature.